Md Ragib Rownak

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Professional Summary

PhD candidate (graduating December 2026) with 5+ years of research experience in AI/ML, optimal control, and physics-based modeling for electrified powertrains and complex engineering systems. Completed a Powertrain and Electrification Controls internship at Cummins Inc., with experience in modeling and simulation for heavy-duty electric vehicles. Skilled in developing end-to-end learning-based solutions, including physics-informed Transformer models and deep reinforcement learning for real-time decision-making, validated against optimal-control benchmarks. Seeking full-time roles in controls, AI/ML engineering, powertrain systems, or applied research.

Education

The Ohio State University — Columbus, OH
Ph.D. in Mechanical Engineering — Jan 2021 – Dec 2026 (expected)
Center for Automotive Research | GPA: 3.59/4.00
Dissertation: Aging-Aware Learning-Based Control for Electrified Powertrains
Advisor: Dr. Qadeer Ahmed

The Ohio State University — Columbus, OH
M.S. in Mechanical Engineering — Dec 2023
Thesis: Uncertainty in Multi-Objective Optimization: Weighted Entropy and Partition-Based Uncertainty Reduction
Advisor: Dr. Carol Smidts

Bangladesh University of Engineering & Technology (BUET) — Dhaka, Bangladesh
B.S. in Mechanical Engineering — Oct 2018
GPA: 3.80/4.00 (Top 8%)

Technical Skills

Languages	Python, MATLAB, C, C++
Controls	Optimal control (Dynamic Programming), energy management strategies, battery aging and capacity-fade estimation, controller hardware-in-the-loop (C-HIL) validation
AI/ML	PyTorch, TensorFlow, Scikit-learn; Transformers, RNN/LSTM, CNNs; Deep Reinforcement Learning (SAC, PPO, TD3, GRPO, DDPG)
Optimization	Mixed-Integer Nonlinear Programming, Markov Decision Processes, Multi-Objective Optimization (NSGA-II), uncertainty quantification, CasADi
Modeling & Sim	MATLAB/Simulink powertrain modeling, physics-based degradation models (battery, electric machine, aftertreatment), LAMMPS
Data & Tools	CAN bus / J1939 log decoding, NumPy, Pandas, Matplotlib, LaTeX
Developer Tools	Agentic AI coding tools (Claude Code, OpenAI Codex); GitHub Copilot; VS Code

Professional Experience

Powertrain Electrification Controls Intern — May 2026 – Aug 2026
Cummins Inc., Columbus, IN

- Analyzed real-world operational data from a fleet of Class 8 commercial trucks to **quantify the hybridization opportunity** for a P2 power-split architecture, identifying adaptive cruise control usage patterns and estimating the recoverable braking energy pool available for regeneration.
- Built a **MATLAB data pipeline** to decode undocumented multi-channel vehicle data logs (J1939/CAN bus signals) from a remote fleet network, enabling automated fleet-wide identification of adaptive cruise control engagement modes and fault-tolerant batch processing over shared network storage.
- Developed a **physics-informed surrogate model** of a Class 8 truck adaptive cruise control decision layer, combining a car-following physics base with a data-driven residual correction; achieved test RMSE of $0.115\,\mathrm{m/s^2}$ ($R^2 = 0.93$), reducing prediction error by nearly half compared to the prior baseline.
- Formulated a **dynamic-programming** framework to compute the globally fuel-optimal energy-management strategy for a Class 8 power-split hybrid, selecting operating modes (EV, hybrid, P2) and engine torque dispatch under charge-sustaining battery constraints; the solution served as the performance benchmark for evaluating real-time hybrid controllers.

Graduate Research Associate, Center for Automotive Research — Jan 2024 – Dec 2025, Aug 2026 – Present
The Ohio State University | Advisor: Dr. Qadeer Ahmed | Industry Partner: Cummins Inc.

- Designed and implemented a **physics-aware deep reinforcement learning** (SAC, PPO, GRPO) framework for hybrid-electric powertrain energy management, achieving fuel consumption within 1.1–4.8% of the dynamic programming optimum while reducing battery degradation by up to 88% and generator lifetime loss by up to 26.3%.
- Developed a **Physics-Informed Transformer Model** for battery capacity fade prediction, reducing estimation error (RSEP) to 2.96% across 17 cells from 5 experimental datasets, outperforming LSTM and standard Transformer baselines.
- Built and validated **physics-based aging models** for battery (SEI growth, active material loss), electric machine (thermal degradation), and aftertreatment catalyst deactivation, integrated into a forward-looking MATLAB/Simulink powertrain simulator for Class 8 heavy-duty series-hybrid vehicles.
- Created a **Dynamic Programming** benchmarking framework to evaluate RL policy quality and generate expert trajectories for accelerated RL training.
- Proposed a **critic-free policy optimization** method (A-GRPO) using Transformer policies and trajectory-level advantage ranking, achieving 2.7× higher terminal constraint feasibility than PPO-Lagrangian on a 3,605-step control task; accepted at IEEE CDC 2026.
- Demonstrated **embedded controller deployment** of trained RL policies on a Speedgoat real-time target in a controller hardware-in-the-loop (C-HIL) testbench (plant: dSPACE SCALEXIO, communication: CAN), reproducing training-time control behavior across 12 deployment runs with less than 1.34% fuel error and 0.06% terminal SOC deviation.
- Delivered quarterly technical results to **Cummins Inc.** through milestone reviews, model validation discussions, and progress reporting.

Graduate Research Associate Jan 2021 – Dec 2023
The Ohio State University | Advisor: Dr. Carol Smidts

- Developed a **VR-based experimental framework** for human reliability assessment under physical security threats at nuclear facilities, integrating biometric data capture with an extended THERP model.
- Designed a **multi-criteria sensor selection framework** using qualitative physical models and NSGA optimization for nuclear facility online monitoring systems.
- Investigated the effect of uncertainty on non-dominated Pareto fronts; derived **analytical probability expressions** for solution dominance and developed a partition-based algorithm using weighted entropy for progressive uncertainty reduction (M.S. thesis).

Graduate Teaching Assistant Jan 2026 – Apr 2026
The Ohio State University

- Led recitation sections and held office hours for ME 3501: Introduction to Engineering Thermodynamics, supporting 100+ students in applying thermodynamic principles to engineering systems.

Lecturer, Department of Mechanical Engineering Jun 2019 – Dec 2020
Bangladesh Army University of Science and Technology (BAUST)

- Taught undergraduate courses (Numerical Analysis, Engineering Mechanics, Mechanics of Materials, Measurement Instrumentation & Quality Control) and conducted lab sessions.
- Supervised student projects including design, construction, and performance analysis of a hydrodynamic tunnel and gravitational water vortex turbine.
- Served as Departmental Program Coordinator, organizing events and facilitating department operations.

Engineering Intern Mar 2018 – Apr 2018
Navana Group, Dhaka, Bangladesh

- Industrial internship in manufacturing processes, quality control, and production line operations.

Publications

First-Author, Current Research (Electrified Powertrains & AI/ML):

[1] **Rownak, M.R.**, Jaleel, W., Hanif, A., Fahim, M.Q., Le, D.D., Anwar, H., Nelson, M., & Ahmed, Q. (2026). Physics-Aware Deep Reinforcement Learning for Energy and Aging Management in Electrified Powertrains. *Accepted, IEEE Transactions on Transportation Electrification.*

[2] **Rownak, M.R.**, Hanif, A., Fahim, M.Q., Le, D.D., Anwar, H., Jaleel, W., Nelson, M., & Ahmed, Q. (2026). Learning Battery Aging Dynamics Using Physics-Informed Transformer. *IEEE Transactions on Transportation Electrification*, 12(2), 3792–3804. doi:10.1109/TTE.2026.3658446.

[3] **Rownak, M.R.**, Bhatti, S.G., & Ahmed, Q. (2026). Ranking-Augmented On-Policy Optimization with Adaptive Advantage-Normalization for Constrained Control. *Accepted, 2026 65th IEEE Conference on Decision and Control (CDC).*

[4] **Rownak, M.R.**, Hanif, A., Ahmed, Q., et al. (2025). Robustness and Sensitivity of Aging Models for Batteries and Electric Machines in Heavy-Duty Electrified Powertrains. In *IEEE ITEC 2025*.

[5] **Rownak, M.R.**, Hanif, A., Ahmed, Q., et al. (2025). Powertrain Components Aging Model Selection for Energy Efficient Vehicles: Selection Strategy and Challenges. *SAE WCX 2025*.

Co-Authored, Current Research:

[6] Jaleel, W., **Rownak, M.R.**, Hanif, A., Bhatti, S.G., & Ahmed, Q. (2026). Sequence Aware SAC Control for Engine Fuel Consumption Optimization in Electrified Powertrain. *Accepted, American Control Conference (ACC) 2026*. arXiv:2508.04874.

First-Author, Human Reliability & Optimization:

[7] **Rownak, M.R.**, Diao, X., & Smidts, C. (2024). Human Reliability Under Physical Security Threats: Modeling and Experimental Design. In *IEEE RAMS 2024*, pp. 1–6.

[8] **Rownak, M.R.** (2023). Uncertainty in Multi-Objective Optimization: Weighted Entropy and Partition-Based Uncertainty Reduction. *Master's Thesis, The Ohio State University.*

Co-Authored, Nuclear Safety & Other:

[9] Diao, X., **Rownak, M.R.**, et al. (2025). A multiple-criteria sensor selection framework based on qualitative physical models. *Advanced Engineering Informatics*, 65, 103228.

[10] Diao, X., **Rownak, M.R.**, & Smidts, C. (2024). Design and Implementation of a Virtual Reality Environment for Human Reliability Assessment. *PBNC 2024*, pp. 716–725.

[11] Diao, X., **Rownak, M.R.**, et al. (2023). Selection Criteria for Optimal Sensor Placement in Online Monitoring Systems. *13th NPIC&HMIT*, pp. 1568–1577.

[12] Olatubosun, S.A., **Rownak, M.R.**, et al. (2022). Human reliability assessment for physical security. In *PSAM 16*, Honolulu, HI.

[13] Hasan, M.N., Paul, S., **Rownak, M.R.**, & Akhter, C. (2019). Thin-film evaporation over biphilic surfaces. *Heat Transfer – Asian Research*, 48, 4283–4300.

[14] Paul, S., **Rownak, M.R.**, & Hasan, M.N. (2020). Molecular dynamics study on thin film evaporation over biphilic surface. *Proc. IMechE Part N.*

Awards & Honors

- Graduate Fellow Award, Dept. of Mechanical and Aerospace Engineering, The Ohio State University (Autumn 2026)
- University Merit Scholarship, Dept. of Mechanical Engineering, BUET (2014)
- Dean's List Award, Dept. of Mechanical Engineering, BUET (2015, 2017)

Professional Service

- Peer Reviewer, *IEEE Transactions on Transportation Electrification* (2026)
- Peer Reviewer, *IEEE Transportation Electrification Conference & Expo (ITEC)* (2026)
- Peer Reviewer, *23rd IFAC World Congress*, Busan, Korea (2026)

References

Dr. Qadeer Ahmed
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Dr. Carol Smidts
Professor, MAE, OSU
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